

AIOps: The Key Enabler for DevOps

The recent pandemic has promoted remote work in a big way, giving a huge boost to DevOps and AIOps. This rapid, wide scale change is creating real concerns in AIOps, DevOps, and IT service management, as organisations seek the best monitoring and incident response solutions for their now distributed enterprises. This article discusses, among other things, the top AIOps tools on GitHub.



AIOps is the application of artificial intelligence in IT operations

Artificial intelligence for IT operations (AIOps) brings together artificial intelligence (AI) with analytics and machine learning (ML) for automation in the identification and resolution of IT operations.

AIOps combines Big Data and machine learning to automate or replace all primary IT operations processes, including availability and performance monitoring, event correlation and analysis, anomaly detection and causality determination, as well as IT service management and automation.

As per Gartner, AIOps refers to 'technology platforms that use machine learning (ML) and data science to solve IT operation problems'. Gartner predicts that the use of AIOps and digital experience monitoring tools to monitor applications and infrastructure will rise from 5 per cent in 2018 to 30 per cent in 2023.

AIOps helps IT operations and DevOps teams to work smarter and faster with the proactive algorithmic analysis of IT data to identify digital service issues and resolve them quickly, before business operations and customers are impacted. With AIOps, Ops teams can understand the immense complexity and quantity of data generated by modern IT environments, and prevent outages, maintain uptime and attain continuous service assurance.

Today's digital era of hybrid infrastructure with heterogeneous commercial off-the-shelf and bespoke (service-based and monolith) apps (that have multiple middleware/ integration platforms with different technology stacks) has mandated IT Ops teams to provide uninterrupted IT service to its end users. This has made IT teams proactive enough to capture and resolve issues before they occur. These teams must deal with constant change while ensuring zero downtime. Seamless operation is the key to grow, compete and thrive.

Earlier, IT teams grew along with technology, from standalone to vertical scaling through distributed computing. The emergence of virtual computing led to a world of microservices and

ephemeral logic with containerised scaling. Nowadays, organisations simply generate too much data for humans to monitor and understand manually or by using legacy tools.

Digital transformation is much more than digitisation of business processes. It brings in all the benefits of digital technologies to make business processes automated, agile and fast with reduced steps and activities, ensuring availability of the required information at a click for the business decision maker. This brings in additional responsibility to ensure that IT operations run proactively for identifying and resolving issues. The key building blocks — scalability, availability, fault tolerance, security, cost-effective, operational excellence — lead us towards IT operations with a DevOps and AIOps strategy.